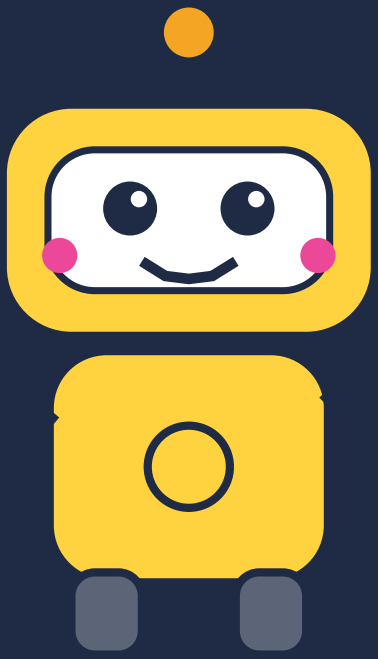
TEACHER BOOK

Think Like a Coder

Teacher book for Class 5

A 5-period plan for every chapter
Hands-on activities, common mistakes and exit checks
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 5 · Follows the 15 chapters of the NCERT Class 5 maths textbook

SAMPLE EDITION · DRAFT

© 2026 Bullseye Knowledge System Limited. All rights reserved.
CodeSchool is a working name. Not an official CBSE or NCERT publication.

What's inside

About this teacher book . . . 3
How a lesson runs . . . 4
Year plan . . . 5
Assessment . . . 6
1. We the Travellers - I . . . 7
2. Fractions . . . 9
3. Angles as Turns . . . 11
4. We the Travellers - II . . . 13
5. Far and Near . . . 15
6. The Dairy Farm . . . 17
7. Shapes and Patterns . . . 19
8. Weight and Capacity . . . 21
9. Coconut Farm . . . 23
10. Symmetrical Designs . . . 25
11. Grandmother's Quilt . . . 27
12. Racing Seconds . . . 29
13. Animal Jumps . . . 31
14. Maps and Locations . . . 33
15. Data Through Pictures . . . 35
Observation journal (photocopy) . . . 37
Term summary (photocopy) . . . 38

© 2026 Bullseye Knowledge System Limited

About this teacher book

This book goes with the *Think Like a Coder* Class 5 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking confidently without being a computer specialist. CBSE suggests about **50 hours** a year for Class 5; the plans here use **75 periods of 40 minutes**, which is about 50 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A 5-period plan for every chapter**: what you do and what students do in each period.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

No computer lab needed

No computers are needed. Every idea is taught with paper, cards, the floor mat and talk. These are the same ideas children will later use in block coding and AI: steps in order, loops, decisions, data and testing. A projector helps for showing pages but is optional.

© 2026 Bullseye Knowledge System Limited

How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

© 2026 Bullseye Knowledge System Limited

Year plan

15 chapters of 5 periods each; revision and the term assessment fit into each chapter's fifth period: **75 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	We the Travellers - I	True, false or can't tell: logic with AND and OR	B4 review	5	Term 1
2	Fractions	Variables that keep score	B5	5	Term 1
3	Angles as Turns	Turtle turns: angles in degrees	B4	5	Term 1
4	We the Travellers - II	Inputs and variables: fuel left	B5	5	Term 1
5	Far and Near	Compare and decide	B5	5	Term 1
6	The Dairy Farm	Counters and running totals	B5	5	Term 1
7	Shapes and Patterns	Pattern generators: rules for any step	B4	5	Term 1
8	Weight and Capacity	Search strategies: halve it to find it fast	B6	5	Term 1
9	Coconut Farm	Input-output rules (function machines)	B5	5	Term 2
10	Symmetrical Designs	Design machines: turning symmetry	B4	5	Term 2
11	Grandmother's Quilt	Tiling and decomposition	B6	5	Term 2
12	Racing Seconds	Timers: variables that change over time	B5	5	Term 2
13	Animal Jumps	Jump rules: loops with conditions	B6	5	Term 2
14	Maps and Locations	Capstone 1: design a treasure-map game	B7	5	Term 2
15	Data Through Pictures	Capstone 2: scores, data and improving the game	B7	5	Term 2

© 2026 Bullseye Knowledge System Limited

Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Variables and tracing	Follows a trace with help	Traces one variable with prompts	Traces counters, totals and inputs alone	Writes programs with variables and inputs
Patterns and rules	Continues a pattern	Finds a simple rule	Uses a rule to jump to any step	Writes rules and function machines
Strategies and decomposition	Searches one by one	Halves with help	Chooses halving or tiling alone	Predicts the number of steps and explains
Logic and decisions	Uses one condition	Uses AND or OR with prompts	Compares and decides with if-else	Designs fair game rules and tests them

✓ How to assess without extra tests

- ✓ **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- ✓ **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- ✓ **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- ✓ **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

© 2026 Bullseye Knowledge System Limited

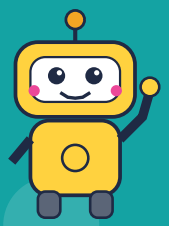
1

TEACHER NOTES · CHAPTER 1 · LEVEL B4 review

We the Travellers - I

Coder skill: True, false or can't tell: logic with AND and OR

Goes with NCERT maths, Class 5, chapter 1



At a glance	
Students will	decide true, false or can't tell, use AND and OR and follow a route program.
Key words	**condition** (something true or false), **AND** (both must be true), **OR** (at least one is true)
Materials	Number cards, a hidden 'mystery number' card, true/false cards, the route grid from the book
Time	5 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *condition*, *AND*, *OR*. Work through the example *A 3-digit number has only odd digits* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Stand up if ...	Run the activity below. Move between groups and note who can explain the idea.	Decide whether AND and OR conditions are true for themselves, judge true, false or can't tell, and write rules.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a checking program using AND. Model the first card or step, then let pairs finish.	Test it with 48, 52 and 51. Which numbers win?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Stand up if ...

1. Call out conditions about the class: 'Stand up if you are wearing a watch AND you have a sister.' Children decide for themselves.
2. Repeat with OR: 'Stand up if you walked to school OR you came by bus.' Compare how many stand.
3. Hold up a hidden number card and say: 'It has 3 digits, all odd.' Ask whether statements are true, false or can't tell.
4. Pairs write one AND rule and one OR rule for a game, and test them with three numbers.

Tip: Ask 'How many stood up for AND, and for OR?' OR is usually more, because only one part needs to be true.

Common mistakes to watch for

- **Treating AND like OR**, standing up when only one part is true.
- **Saying 'false' when the answer is 'can't tell'.** If it could go either way, we cannot tell yet.
- **Missing the edge case**: 'more than 50' does not include 50.

Different paces

Support: Use one condition at a time before joining two.
Stretch: Write a rule that uses AND and OR together, and find a number that makes it true and one that makes it false.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Rule: win if the number is even AND more than 50. Does 50 win? (Answer: no: 50 is not more than 50)

2. Rule: prize if the score is more than 8 OR you finish first. Riya scored 9 and finished third. Does she get a prize? (Answer: yes)

Note in the journal: Beginning (needs help to decide true, false or can't tell); **Developing** (can decide true, false or can't tell with prompts or a partner); **Secure** (can decide true, false or can't tell and use AND and OR alone); **Extending** (can follow a route program and explain why).

Answer key

1. a) It is even
2. b) no: both must be true
3. a) yes: one is enough with OR
4. 326
5. a) C3 (facing South)
Build it: Only 52 (48 is not more than 50; 51 is not even)
Challenge: 20,468 (0 cannot come first)

© 2026 Bullseye Knowledge System Limited

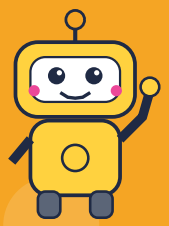
2

TEACHER NOTES · CHAPTER 2 · LEVEL B5

Fractions

Coder skill: Variables that keep score

Goes with NCERT maths, Class 5, chapter 2



At a glance	
Students will	find a fraction of a number, trace a variable and compare fractions.
Key words	**variable** (a named box for a number), **set** (put a value in the box), **change** (add or take away from the box)
Materials	A box labelled *sweets*, 24 counters, fraction strips, trace tables
Time	5 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *variable*, *set*, *change*. Work through the example *Trace the variable 'slices'* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The shrinking box	Run the activity below. Move between groups and note who can explain the idea.	Change a variable with fraction steps, record a trace table, and write fraction programs for a partner.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a program with a variable called **sweets**. Model the first card or step, then let pairs finish.	Trace it in a table. What does the program say at the end?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: The shrinking box

1. Set the variable: put 24 counters in the box labelled *sweets*.
2. Read the program: *give away half; give away a quarter of what is left; add 4*. A child changes the counters at each step.
3. The class fills a trace table: step, what happens, sweets (24, 12, 9, 13).
4. Pairs write their own 3-step program with fractions and trace each other's.

Tip: Ask 'half of what?' at every step. The fraction is always of the value *now*, not of the start.

Common mistakes to watch for

- **Taking the fraction of the starting value** instead of the current value.
- **Skipping rows in the trace table** and losing track of the value.
- **Comparing fractions by the bottom number only**: 1/3 is bigger than 1/4.

Different paces

Support: Use halves only, with 16 counters.
Stretch: Find a starting number that gives a whole number at every step of 'give away half, then a third of what is left'.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Set water to 16. Use half. Then use half of what is left. What is water now? (Answer: 4)

2. Which is bigger: 2/3 or 3/5? (Answer: 2/3 (10/15 against 9/15))

Note in the journal: Beginning (needs help to find a fraction of a number); **Developing** (can find a fraction of a number with prompts or a partner); **Secure** (can find a fraction of a number and trace a variable alone); **Extending** (can compare fractions and explain why).

Answer key

1. 12 and 8
2. a) 3/4
3. 5 litres
4. b) 1/2
5. 3
Build it: 20, then 10, then 5: it says 5
Challenge: 80 litres (1/4 is 20 litres)

© 2026 Bullseye Knowledge System Limited

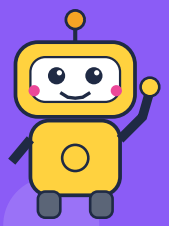
3

TEACHER NOTES · CHAPTER 3 · LEVEL B4

Angles as Turns

Coder skill: Turtle turns: angles in degrees

Goes with NCERT maths, Class 5, chapter 3



At a glance	
Students will	measure turns in degrees, turn clockwise and anticlockwise and draw shapes with turns that add to 360°.
Key words	**degree** (a unit for turning), **right angle** (a 90 degree turn), **clockwise** (the way clock hands turn)
Materials	Chalk or tape for a playground, turn cards (90°, 120°, 60°), paper circles folded into quarters
Time	5 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *degree*, *right angle*, *clockwise*. Work through the example *Drawing a square and a triangle* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The human turtle	Run the activity below. Move between groups and note who can explain the idea.	Walk turtle programs, mark the shapes, and check that the turns in a closed shape add up to 360°.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a turtle program that draws a hexagon with 6 equal sides. Model the first card or step, then let pairs finish.	Check: do your turns add up to 360°?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: The human turtle

1. Fold a paper circle into quarters to show 90°, 180° and 360° turns.
2. A child is the turtle. The class gives *repeat 4 times: walk 4 steps, turn right 90°*. Mark the path with chalk.
3. Try *repeat 3 times: walk 4 steps, turn right 120°*. Ask: 'What did all the turns add up to?' (360°)
4. Pairs work out the turn for a hexagon (60°) and test it on the playground.

Tip: The turtle turns on the spot. Have the class say the running total of degrees after each turn.

Common mistakes to watch for

- **Using the angle inside the shape as the turn.** A triangle's turtle turn is 120°, not 60°.
- **Mixing up clockwise and anticlockwise.** Clockwise is the way clock hands move.
- **Not checking the total.** The turns of a closed shape add up to 360°.

Different paces

Support: Use only 90° turns to draw squares and rectangles.
Stretch: Work out the turn for a regular octagon (45°) and write its program.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What turn does a turtle make at each corner of a regular hexagon? (Answer: 60°)

2. You face West and turn 90° clockwise. Which way do you face? (Answer: North)

Note in the journal: Beginning (needs help to measure turns in degrees); **Developing** (can measure turns in degrees with prompts or a partner); **Secure** (can measure turns in degrees and turn clockwise and anticlockwise alone); **Extending** (can draw shapes with turns that add to 360° and explain why).

Answer key

1. 180
2. a) East
3. c) West
4. b) triangle
5. b) 45°
Build it: Repeat 6 times: forward 30, turn right 60° (6 x 60 = 360)
Challenge: 72° (360 / 5)

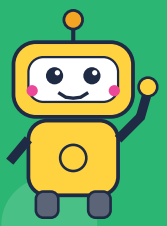
© 2026 Bullseye Knowledge System Limited

TEACHER NOTES · CHAPTER 4 · LEVEL B5

4 We the Travellers - II

Coder skill: Inputs and variables: fuel left

Goes with NCERT maths, Class 5, chapter 4



At a glance	
Students will	use an input in a program, update a variable and solve travel problems step by step.
Key words	**input** (information given to a program), **output** (what the program gives back), **update** (change a variable's value)
Materials	Fuel cards (a car with 40 L), distance input cards, a board for the fuel variable
Time	5 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *input*, *output*, *update*. Work through the example *How much fuel is left?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The fuel gauge game	Run the activity below. Move between groups and note who can explain the idea.	Update a fuel variable from distance inputs, apply a warning rule, and name inputs and outputs.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a fuel program that uses an input. Model the first card or step, then let pairs finish.	Run it with 100 km, then 450 km. What does it say each time?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The fuel gauge game

1. A child is the car, holding a fuel card set to 40 L. The rule: 8 L for every 100 km.
2. Others draw input cards with distances: 100 km, 50 km, 200 km. After each input, the class works out the fuel used and updates the fuel card.
3. Add the warning rule: *if fuel is less than 5, say 'Fill up soon!'*. Keep going until it fires.
4. Ask: 'Which part is the input, which is the variable, and which is the output?'

Tip: Write 'input', 'variable' and 'output' as labels on the board and point to each as it changes.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Subtracting the distance instead of the fuel used.** Work out litres first.
- **Forgetting to update the variable** after each input.
- **Mixing up input and output.** The input comes in; the output is what the program says.

Different paces

Support: Use only 100 km inputs (8 L each).
Stretch: What is the longest trip on a full 40 L tank? (500 km) After how many km does the warning come on? (after more than 437.5 km, when over 35 L is used)

Exit check and what to note

Two quick questions at the end of the chapter:

1. Fuel is 40 L and the car uses 8 L per 100 km. The input is 250 km. How much fuel is left? (Answer: 20 L)

2. In the fuel program, what is the input? (Answer: the distance travelled, given while the program runs)

Note in the journal: Beginning (needs help to use an input in a program); **Developing** (can use an input in a program with prompts or a partner); **Secure** (can use an input in a program and update a variable alone); **Extending** (can solve travel problems step by step and explain why).

Answer key

1. 24 litres
2. 3 badges and 1 sticker
3. b) Teen or adult
4. 470 km
5. 15
Build it: 100 km: fuel 32, no warning; 450 km: fuel 4, says 'Fill up soon!'
Challenge: 500 km

© 2026 Bullseye Knowledge System Limited

5

TEACHER NOTES · CHAPTER 5 · LEVEL B5

Far and Near

Coder skill: Compare and decide

Goes with NCERT maths, Class 5, chapter 5



At a glance	
Students will	compare distances, choose with an if-else rule and find the nearest with a variable.
Key words	**compare** (check which is bigger), **less than (<)** (smaller than), **kilometre** (1,000 metres)
Materials	Shop cards with distances face down, a board for the variable *nearest*, a local map
Time	5 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *compare*, *less than (<)*, *kilometre*. Work through the example *Walk, cycle or bus?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Nearest shop hunt	Run the activity below. Move between groups and note who can explain the idea.	Keep a 'nearest so far' variable while turning over cards, and update it with a comparison.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a 'nearest shop' program that uses a variable called **nearest**. Model the first card or step, then let pairs finish.	Trace it with shops at 400 m, 250 m and 600 m. What is nearest at the end?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Nearest shop hunt

1. Place 5 shop cards face down, each with a distance in metres or kilometres.
2. Turn the first card: *set nearest to its distance*. Write it on the board.
3. Turn each next card and compare: *if it is less than nearest, then update nearest*.
4. At the end, ask: 'Did we need to compare every card? Why?' (Yes: any card could be the nearest.)

Tip: Mix units (1 km 200 m and 900 m) so children must convert before comparing.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Comparing without converting units**: 1 km is 1,000 m.
- **Updating the variable when the new distance is bigger.** Only update when it is smaller.
- **Stopping early** before checking every place.

Different paces

Support: Use only metres and 3 cards.
Stretch: Change the program to find the farthest shop, and the second nearest.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Shops are at 700 m, 350 m and 500 m. What is *nearest* after each check? (Answer: 700, then 350, then 350)

2. Is 1 km more or less than 900 m? (Answer: more: 1 km is 1,000 m)

Note in the journal: Beginning (needs help to compare distances); **Developing** (can compare distances with prompts or a partner); **Secure** (can compare distances and choose with an if-else rule alone); **Extending** (can find the nearest with a variable and explain why).

Answer key

1. 2,300 m
2. b) 900 m
3. b) walk (3 is not more than 3)
4. b) no
5. 980 m; 1 km 50 m; 1,200 m
Build it: 400, then 250, then stays 250: nearest is 250 m
Challenge: 85 minutes faster (walking takes 1 hour 40 minutes; the bus takes 15 minutes)

© 2026 Bullseye Knowledge System Limited

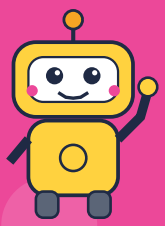
6

TEACHER NOTES · CHAPTER 6 · LEVEL B5

The Dairy Farm

Coder skill: Counters and running totals

Goes with NCERT maths, Class 5, chapter 6



At a glance	
Students will	use a counter, keep a running total and read a picture graph with half symbols.
Key words	**counter** (goes up by 1 each time), **total** (adds up amounts), **litre** (1,000 millilitres)
Materials	Cow cards with litres of milk, a board with *counter* and *total*, a picture graph frame
Time	5 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *counter*, *total*, *litre*. Work through the example *Milk from 5 cows* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The milk collection centre	Run the activity below. Move between groups and note who can explain the idea.	Keep a counter and a running total as cows arrive, then show the milk in a picture graph.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a milk-counting program with a counter and a total. Model the first card or step, then let pairs finish.	Why must total start at 0 and not at 10?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The milk collection centre

1. Children are cows, each holding a card with litres of milk (such as 5, 7, 6, 4).
2. The collector sets *counter* = *0* and *total* = *0* on the board.
3. Each cow arrives: *counter* = *counter* + *1*, *total* = *total* + *milk*. Update the board each time.
4. Show the result as a picture graph where one symbol = 2 litres, using half symbols.

Tip: Ask 'What would happen if total started at 10?' to show why we set it to 0 before the loop.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Not starting the total at 0.** The first amount is then added to the wrong value.
- **Mixing up the counter and the total**: the counter goes up by 1, the total by the amount.
- **Reading half symbols as whole ones** in the picture graph.

Different paces

Support: Use 3 cows with small amounts.
Stretch: Add an average: total ÷ counter. What is the average milk per cow?

Exit check and what to note

Two quick questions at the end of the chapter:

1. Counter and total start at 0. Cows give 6, 5 and 7 litres. What are counter and total? (Answer: counter 3, total 18)

2. Why must total start at 0? (Answer: otherwise the first amount is added to a wrong starting value)

Note in the journal: Beginning (needs help to use a counter); **Developing** (can use a counter with prompts or a partner); **Secure** (can use a counter and keep a running total alone); **Extending** (can read a picture graph with half symbols and explain why).

Answer key

1. 50
2. 25 litres
3. 3
4. 2,500 ml
5. 80
Build it: If it starts at 10, every answer is 10 too big
Challenge: Rs 14,000

© 2026 Bullseye Knowledge System Limited

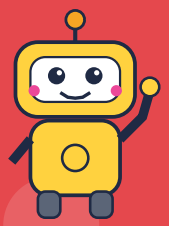
7

TEACHER NOTES · CHAPTER 7 · LEVEL B4

Shapes and Patterns

Coder skill: Pattern generators: rules for any step

Goes with NCERT maths, Class 5, chapter 7



At a glance	
Students will	continue a growing pattern, find a pattern's rule and use a rule to jump to any step.
Key words	**rule** (tells how a pattern grows), **step** (one stage of a pattern), **generator** (a rule that makes the pattern)
Materials	Matchsticks or straws, squared paper, pattern tables
Time	5 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *rule*, *step*, *generator*. Work through the example *Matchstick squares in a row* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The matchstick factory	Run the activity below. Move between groups and note who can explain the idea.	Build a growing pattern, record it in a table, find its rule and use it to jump to step 20.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Turn this rule into a table: **tiles = step x 2 + 1**. Model the first card or step, then let pairs finish.	Fill the table, then draw step 3.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The matchstick factory

1. Groups build steps 1 to 4 of matchstick squares in a row and record the sticks in a table.
2. Ask: 'What changes from step to step?' (3 more sticks each time.)
3. Groups write the rule *sticks = 3 x squares + 1* and predict step 20 without building it.
4. Swap rules with another group: each group builds a new pattern and the other finds its rule.

Tip: Ask 'Where does the +1 come from?' Pointing to the first stick in the row makes the rule meaningful.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Only adding on** (3, 3, 3, ...) instead of writing a rule for any step.
- **Forgetting the extra part** of the rule, such as the + 1.
- **Testing the rule on one step only.** Check it against every step in the table.

Different paces

Support: Start with a pattern that grows by 2 and begins at 2 (2, 4, 6, ...).
Stretch: Find the rule for triangles in a row (3, 5, 7 sticks) and the step that needs exactly 41 sticks (step 20).

Exit check and what to note

Two quick questions at the end of the chapter:

1. Rule: tiles = step x 2 + 1. How many tiles in step 10? (Answer: 21)

2. A pattern goes 5, 9, 13, 17. What is its rule? (Answer: step x 4 + 1)

Note in the journal: Beginning (needs help to continue a growing pattern); **Developing** (can continue a growing pattern with prompts or a partner); **Secure** (can continue a growing pattern and find a pattern's rule alone); **Extending** (can use a rule to jump to any step and explain why).

Answer key

1. 16
2. 61
3. 16 (4 x 4)
4. b) step x 3
5. a) triangle (17 = 4 x 4 + 1)
Build it: Steps 1-5: 3, 5, 7, 9, 11 tiles
Challenge: Step 20 uses 61 sticks; step 33 uses 100 (3 x 33 + 1 = 100)

© 2026 Bullseye Knowledge System Limited

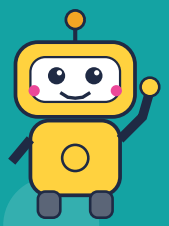
8

TEACHER NOTES · CHAPTER 8 · LEVEL B6

Weight and Capacity

Coder skill: Search strategies: halve it to find it fast

Goes with NCERT maths, Class 5, chapter 8



At a glance	
Students will	search one by one, search by halving and explain which strategy is faster.
Key words	**linear search** (check one by one), **halving search** (check the middle, drop half), **strategy** (a plan for solving)
Materials	Number strip 1-32 (floor or board), 8 coin counters, a simple balance or its picture, search code cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 8 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *linear search*, *halving search*, *strategy*. Work through the example *Guess my number (1 to 32)* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Guess my number	Run the activity below. Move between groups and note who can explain the idea.	Compare one-by-one search with halving, count the guesses, and find a heavy coin by halving.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write the 'guess my number' program using halving. Model the first card or step, then let pairs finish.	Play with a friend. Count your guesses. Is it ever more than 6?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Guess my number

1. Think of a number from 1 to 32. Let the class guess 1, 2, 3, ... in order and count the guesses.
2. Play again, but now guess the middle each time and grey out the half that is ruled out.
3. Ask: 'How many numbers did one guess rule out? Why is the middle the best guess?'
4. Show 8 coins, one secretly heavier. Pairs plan weighings that halve the coins each time and find it in 3 weighings.

Tip: Shade each ruled-out half on the strip so children see the range shrink.

Common mistakes to watch for

- **Guessing the middle of the whole strip again** instead of the middle of what is left.
- **Off by one:** forgetting that the guessed number itself is also ruled out.
- **Thinking halving always wins.** For 3 cards, checking one by one is just as quick.

Different paces

Support: Use the range 1-16 and grey out each half on the strip.
Stretch: How many guesses for 1-100? For 1-1,000? (about 7 and 10)

Exit check and what to note

Two quick questions at the end of the chapter:

1. For 1-64, at most how many halving questions are needed? (Answer: 6)

2. One of 8 coins is heavier. How many weighings on a balance? (Answer: 3)

Note in the journal: Beginning (needs help to search one by one); **Developing** (can search one by one with prompts or a partner); **Secure** (can search one by one and search by halving alone); **Extending** (can explain which strategy is faster and explain why).

Answer key

1. b) 3 (4 vs 4, then 2 vs 2, then 1 vs 1)
2. 20
3. 4
4. 17 to 23
5. 8
Build it: No: halving finds any number from 1 to 32 in at most 6 guesses (5 yes/no splits)
Challenge: 2: weigh 3 against 3; the heavy group (or the 3 left out) is known; then weigh 1 against 1 from that group

© 2026 Bullseye Knowledge System Limited

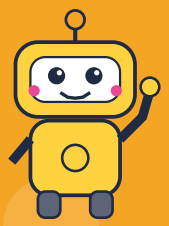
9

TEACHER NOTES · CHAPTER 9 · LEVEL B5

Coconut Farm

Coder skill: Input-output rules (function machines)

Goes with NCERT maths, Class 5, chapter 9



At a glance	
Students will	use an input-output rule, find a hidden rule and chain two machines.
Key words	**function machine** (input, rule, output), **rule** (what the machine does), **chain** (join machines one after another)
Materials	A cardboard box as a 'machine', input and output cards, rule cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 9 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *function machine*, *rule*, *chain*. Work through the example *The harvest calculator* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The human function machine	Run the activity below. Move between groups and note who can explain the idea.	Feed inputs to a hidden-rule machine, record outputs, find the rule, and chain two machines.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design your own two-step machine and test it on a friend. Model the first card or step, then let pairs finish.	Give your friend three inputs and outputs. Can they find your rule?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: The human function machine

1. A child sits behind the box with a secret rule card, such as *x 2 + 1*.
2. The class feeds in input cards (3, 5, 10); the machine passes back outputs (7, 11, 21). Record them in a table.
3. After three inputs, pairs guess the rule and test it on a new input.
4. Chain two machines: the output of the first becomes the input of the second.

Tip: Choose inputs that make the rule easy to see, such as 0 and 1, before harder ones.

Common mistakes to watch for

- **Guessing a rule from one input only.** Test it on every pair in the table.
- **Doing the steps in the wrong order**: x 2 then + 1 is not the same as + 1 then x 2.
- **Forgetting that a chain uses the first output** as the second input.

Different paces

Support: Use one-step rules such as + 5 or x 3.
Stretch: Find two different chains that turn 4 into 20.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Input 4 gives 13, and input 6 gives 19. What is the rule? (Answer: x 3 + 1)

2. Machine A adds 5; machine B doubles. What comes out if 3 goes through A, then B? (Answer: 16)

Note in the journal: Beginning (needs help to use an input-output rule); **Developing** (can use an input-output rule with prompts or a partner); **Secure** (can use an input-output rule and find a hidden rule alone); **Extending** (can chain two machines and explain why).

Answer key

1. 25
2. b) x 5
3. 12
4. 750
5. 62 bags, 6 left over
Build it: Open answer
Challenge: 17; no: + 3 first gives 20, so the order matters

© 2026 Bullseye Knowledge System Limited

10

TEACHER NOTES · CHAPTER 10 · LEVEL B4

Symmetrical Designs

Coder skill: Design machines: turning symmetry

Goes with NCERT maths, Class 5, chapter 10



At a glance	
Students will	find rotational symmetry, work out the turn for n petals and write a design program.
Key words	**rotational symmetry** (looks the same after a turn), **order** (how many matching positions), **reflection** (a mirror image)
Materials	Paper circles, tracing paper, a turn guide (circle folded into equal parts), colours
Time	5 periods of 40 minutes
Link	Teach alongside chapter 10 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *rotational symmetry*, *order*, *reflection*. Work through the example *A 4-petal flower* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Rangoli turns	Run the activity below. Move between groups and note who can explain the idea.	Draw petal designs with repeat-and-turn programs and check their order of rotational symmetry.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a rangoli program for a design with 5 petals. Model the first card or step, then let pairs finish.	Draw the design and colour it so it still has rotational symmetry.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Rangoli turns

1. Fold a paper circle into 4 equal parts. Draw a petal along one fold, turn the paper to the next fold, and repeat.
2. Write the program: *repeat 4 times: draw a petal, turn 90°.*
3. Groups make 3-, 6- and 8-petal designs, working out each turn as 360 ÷ petals.
4. Check the order of rotational symmetry by turning a tracing of each design over the original.

Tip: Colour carefully: one wrong colour can break the rotational symmetry.

Common mistakes to watch for

- **Using the wrong turn**, such as 60° for 5 petals. Work out 360 ÷ 5 = 72°.
- **Counting the starting position twice** when finding the order.
- **Mixing up rotational symmetry and lines of symmetry.** Turning is not folding.

Different paces

Support: Use 4 petals with the folded circle as a guide.
Stretch: Design a pattern with rotational symmetry of order 4 but no lines of symmetry (like a pinwheel).

Exit check and what to note

Two quick questions at the end of the chapter:

1. What turn is needed for a design with 8 petals? (Answer: 45°)

2. A design matches itself 3 times in a full turn. What is its order? (Answer: 3)

Note in the journal: Beginning (needs help to find rotational symmetry); **Developing** (can find rotational symmetry with prompts or a partner); **Secure** (can find rotational symmetry and work out the turn for n petals alone); **Extending** (can write a design program and explain why).

Answer key

1. 6
2. c) 60°
3. d) N, S and Z
4. 3
5. 45°
Build it: Repeat 5 times: draw a petal, turn 72°
Challenge: 9 (9 x 40 = 360)

© 2026 Bullseye Knowledge System Limited

TEACHER NOTES · CHAPTER 11 · LEVEL B6

11 Grandmother's Quilt

Coder skill: Tiling and decomposition

Goes with NCERT maths, Class 5, chapter 11



At a glance	
Students will	find the tile that repeats, count area, including halves and plan a design by decomposition.
Key words	**tile** (a part that repeats), **area** (how many square units cover a shape), **decompose** (break into parts)
Materials	Squared paper, scissors, coloured pencils, a sample patchwork or quilt picture
Time	5 periods of 40 minutes
Link	Teach alongside chapter 11 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *tile*, *area*, *decompose*. Work through the example *Grandmother's quilt* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Quilt workshop	Run the activity below. Move between groups and note who can explain the idea.	Design a tile, repeat it with a nested loop, and count area including half squares.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design a 2 x 2 tile, then write a nested loop to repeat it across a 4 x 4 quilt. Model the first card or step, then let pairs finish.	Draw your quilt on squared paper. How many tiles did you place?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Quilt workshop

1. Show a quilt picture and ask: 'What is the smallest tile that repeats?'
2. Each child designs a 2 by 2 tile on squared paper, using full squares and half-triangles.
3. Groups write a nested loop to repeat the tile across a 4 by 4 quilt and draw it.
4. Work out the shaded area of one tile, counting each half-triangle as half a square unit.

Tip: Cut out one tile and slide it across the quilt to check the nested loop.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Counting a half-triangle as a whole square.**
- **Rotating the tile by accident** when copying it, so the pattern changes.
- **Wrong loop counts**: a 4 by 4 quilt of 2 by 2 tiles needs 2 rows of 2 tiles.

Different paces

Support: Use full squares only in the tile.
Stretch: Make a quilt where every second tile is turned a quarter turn. Write the program.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A tile has 2 full squares and 4 half-triangles shaded. What is the shaded area? (Answer: 4 square units)

2. How many 2 by 2 tiles cover a 6 by 6 quilt? (Answer: 9)

Note in the journal: Beginning (needs help to find the tile that repeats); **Developing** (can find the tile that repeats with prompts or a partner); **Secure** (can find the tile that repeats and count area, including halves alone); **Extending** (can plan a design by decomposition and explain why).

Answer key

1. 4
2. 8
3. 4 square units (8 halves)
4. 24
5. 6
Build it: 4 tiles
Challenge: 16 (there are 16 tiles)

© 2026 Bullseye Knowledge System Limited

TEACHER NOTES · CHAPTER 12 · LEVEL B5

12

Racing Seconds

Coder skill: Timers: variables that change over time

Goes with NCERT maths, Class 5, chapter 12



At a glance	
Students will	change minutes into seconds, compare race times and write a countdown.
Key words	**timer** (counts seconds), **countdown** (counts back to zero), **second** (60 make a minute)
Materials	Stopwatch or phone timer, a 10 m track, timer cards, results table
Time	5 periods of 40 minutes
Link	Teach alongside chapter 12 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *timer*, *countdown*, *second*. Work through the example *Who won the race?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Stopwatch races	Run the activity below. Move between groups and note who can explain the idea.	Time a short race, convert and sort times, and act out a countdown variable.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a countdown program for a race start. Model the first card or step, then let pairs finish.	Act it out: one friend is the timer and says each number. Does 'Go!' come after 0?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Stopwatch races

1. In pairs, one walks 10 m heel to toe while the other times it; then swap.
2. Record all times in seconds on the board, converting any minutes (1 min 5 s = 65 s).
3. Sort the times: the smallest time wins.
4. A child acts as a countdown timer from 10 to 0 and then says 'Go!' to start the next race.

Tip: Practise 'timer = timer - 1' aloud with the countdown so the variable idea is clear.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Thinking the biggest time wins.** In a race, the smallest time wins.
- **Converting minutes as 100 seconds.** 1 minute = 60 seconds.
- **Saying 'Go' at 1** instead of after 0.

Different paces

Support: Use times in seconds only, under one minute.
Stretch: Write a program that times a race and says 'New record!' if the time is less than the best so far.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many seconds are there in 2 minutes 15 seconds? (Answer: 135)

2. Race times are 0:48, 0:51 and 0:45. Which time wins? (Answer: 0:45)

Note in the journal: Beginning (needs help to change minutes into seconds); **Developing** (can change minutes into seconds with prompts or a partner); **Secure** (can change minutes into seconds and compare race times alone); **Extending** (can write a countdown and explain why).

Answer key

1. 135
2. a) 48 s
3. 180
4. b) 1 minute
5. 10
Build it: Yes: the loop stops at 0, then the program says 'Go!'
Challenge: 80 seconds (1 minute 20 seconds)

© 2026 Bullseye Knowledge System Limited

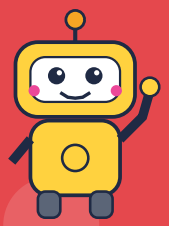
13

TEACHER NOTES · CHAPTER 13 · LEVEL B6

Animal Jumps

Coder skill: Jump rules: loops with conditions

Goes with NCERT maths, Class 5, chapter 13



At a glance	
Students will	skip count with jumps, find common landings and use a loop with a condition.
Key words	**multiple** (a number in a times table), **common multiple** (in both times tables), **repeat until** (loop until a condition is true)
Materials	Chalk number line 0-30, two colours of chalk or sticky notes, animal cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 13 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *multiple*, *common multiple*, *repeat until*. Work through the example *Frog (3) and rabbit (4)* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Frog and rabbit jumps	Run the activity below. Move between groups and note who can explain the idea.	Mark animal landings on a floor line, find common multiples, and write a loop with a condition.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a jump game program with a condition inside the loop. Model the first card or step, then let pairs finish.	On which positions does the program say 'Splash!'?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Frog and rabbit jumps

1. Draw a number line 0-30 on the floor. A 'frog' jumps 3 at a time; mark its landings in one colour.
2. A 'rabbit' jumps 4 at a time; mark its landings in another colour.
3. Ask: 'Where did both land?' (12 and 24, the common multiples.)
4. Write the loop: *repeat until position > 30: jump 3; if position is a multiple of 4, say 'Meet!'*

Tip: Let the class say each landing aloud. Skip counting together makes the multiples stick.

Common mistakes to watch for

- **Including 0 or starting at the jump size wrongly.** The first landing is the jump size.
- **Missing a common landing** because one list was not continued far enough.
- **Mixing up multiples and factors.** Multiples are the landings; they get bigger.

Different paces

Support: Use jumps of 2 and 5 on a line to 20.
Stretch: Three animals jump 2, 3 and 4. Where do all three first meet? (12)

Exit check and what to note

Two quick questions at the end of the chapter:

1. A frog jumps 2 at a time and a kangaroo 5 at a time. Where do they first both land after 0? (Answer: 10)

2. List the landings of an animal that jumps 3 at a time, up to 15. (Answer: 3, 6, 9, 12, 15)

Note in the journal: Beginning (needs help to skip count with jumps); **Developing** (can skip count with jumps with prompts or a partner); **Secure** (can skip count with jumps and find common landings alone); **Extending** (can use a loop with a condition and explain why).

Answer key

1. a) yes (7 jumps)
2. 12 and 24
3. 24
4. b) no (30 is not a multiple of 4)
5. 70
Build it: 15 and 30
Challenge: 24

© 2026 Bullseye Knowledge System Limited

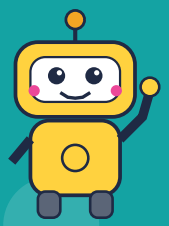
14

TEACHER NOTES · CHAPTER 14 · LEVEL B7

Maps and Locations

Coder skill: Capstone 1: design a treasure-map game

Goes with NCERT maths, Class 5, chapter 14



At a glance	
Students will	use grid references and directions, use a map scale and design a game with clear rules.
Key words	**grid reference** (a letter and number for a square), **scale** (map distance to real distance), **game design** (planning goal, rules and winning)
Materials	Squared paper (5 by 6 grids), compass rose cards, arrow cards, coloured pencils
Time	5 periods of 40 minutes
Link	Teach alongside chapter 14 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *grid reference*, *scale*, *game design*. Work through the example *The treasure map* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Game-design studio	Run the activity below. Move between groups and note who can explain the idea.	Design a treasure-map game with clear rules, test another group's game, and fix problems found in testing.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design your own treasure-map game on a 5 x 6 grid. Model the first card or step, then let pairs finish.	Draw your map and write your rules. Test it: can the treasure always be reached?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Game-design studio

1. Groups draw a 5 by 6 map with a start, a treasure and some walls, and label the grid references.
2. They write the rules: how to move, what counts as a win, what happens at a wall.
3. Another group plays the game with arrow cards and notes anything confusing or impossible.
4. Designers fix one problem and test again. Keep the maps for chapter 15.

Tip: Ask testers to try to 'break' the game: can they reach the treasure? Is any rule unclear? Testing is part of design.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Mixing up the grid reference order.** Agree: letter (column) first, then number (row).
- **Treasure that cannot be reached** because walls block every route.
- **Rules that do not say what happens** at walls or at the edge of the map.

Different paces

Support: Use a 4 by 4 grid with no walls.
Stretch: Add a key that must be collected before the treasure, and make the shortest route exactly 9 moves.

Exit check and what to note

Two quick questions at the end of the chapter:

1. The map scale is 1 square = 100 m. A route is 7 squares long. How far is it? (Answer: 700 m)

2. Robo is at C3 and moves 2 squares North. What is the new grid reference? (Answer: C5)

Note in the journal: Beginning (needs help to use grid references and directions); **Developing** (can use grid references and directions with prompts or a partner); **Secure** (can use grid references and directions and use a map scale alone); **Extending** (can design a game with clear rules and explain why).

Answer key

1. b) a tree
2. E5
3. 500 m
4. b) South-West
5. 9
Build it: Open answer
Challenge: Open answer

© 2026 Bullseye Knowledge System Limited

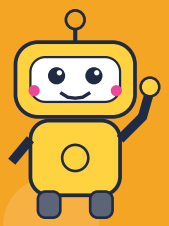
TEACHER NOTES · CHAPTER 15 · LEVEL B7

15

Data Through Pictures

Coder skill: Capstone 2: scores, data and improving the game

Goes with NCERT maths, Class 5, chapter 15



At a glance	
Students will	read picture graphs and bar graphs, keep score with a variable and use data to improve a game.
Key words	**score** (a variable for points), **pictograph** (a graph with pictures), **test** (try it to find problems)
Materials	The game maps from chapter 14, score sheets, squared paper for graphs
Time	5 periods of 40 minutes
Link	Teach alongside chapter 15 of the NCERT Class 5 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *score*, *pictograph*, *test*. Work through the example *Scores from the treasure game* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Play-test day	Run the activity below. Move between groups and note who can explain the idea.	Add a score, collect play-test data, graph it, and improve the game based on the data.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Finish your game from chapter 14 by adding a score. Model the first card or step, then let pairs finish.	Play with 4 friends, draw a graph of the scores and write one change that would make the game better.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Play-test day

1. Groups add a score rule to their game, such as +10 for the treasure and -2 for each wall bump.
2. Other groups play; designers record each player's moves, bumps and score in a data table.
3. Groups draw a bar graph or picture graph of the scores.
4. From the data, each group writes one change to make the game fairer or more fun, makes it, and tests again.

Tip: Ask 'What does your data say?' before 'What do you think?' The point is to let evidence drive the change.

© 2026 Bullseye Knowledge System Limited

Common mistakes to watch for

- **Changing the game without looking at the data.** Point to the evidence first.
- **Graphs without titles or labels.**
- **Ties with no tie-break rule.** Decide one before playing.

Different paces

Support: Record only the number of moves, not a score.
Stretch: Compare the average score before and after the change. Did the change help?

Exit check and what to note

Two quick questions at the end of the chapter:

1. Score rule: +10 for each treasure, -2 for each wall bump. A player finds 2 treasures and bumps 3 walls. What is the score? (Answer: 14)

2. In a picture graph, one symbol = 4 points. A row has 3 and a half symbols. How many points? (Answer: 14)

Note in the journal: Beginning (needs help to read picture graphs and bar graphs); **Developing** (can read picture graphs and bar graphs with prompts or a partner); **Secure** (can read picture graphs and bar graphs and keep score with a variable alone); **Extending** (can use data to improve a game and explain why).

Answer key

1. 8
2. 5
3. 26
4. b) Dev
5. 5
Build it: Open answer
Challenge: Open answer, for example: give bonus points for the shortest route so players plan instead of rushing

© 2026 Bullseye Knowledge System Limited

Observation journal

Class 5 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Variables	Patterns	Strategies	Logic	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

© 2026 Bullseye Knowledge System Limited

Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Variables, inputs and tracing	B D S E	
Patterns, rules and functions	B D S E	
Search and decomposition strategies	B D S E	
Game design and data (capstone)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

© 2026 Bullseye Knowledge System Limited